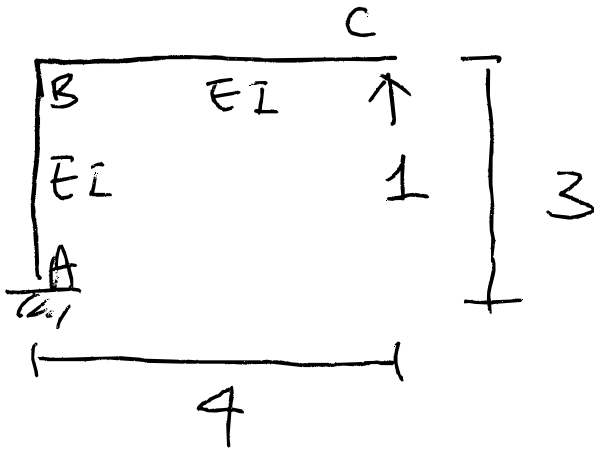
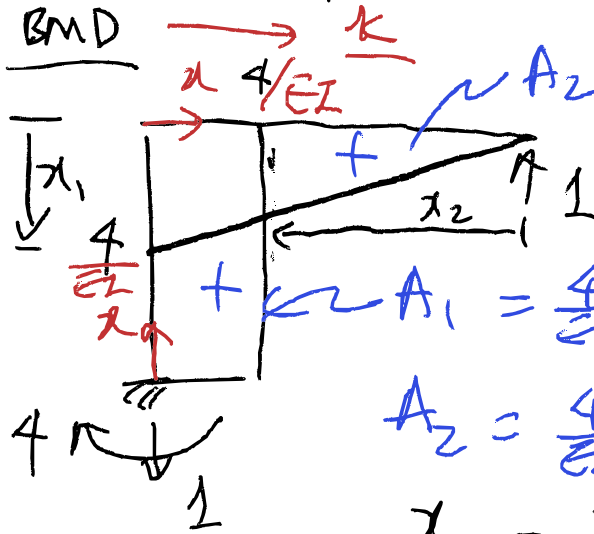


Example 1



$$SI = 0$$



$$A_1 = \frac{4}{EI} \times 3 = \frac{12}{EI}$$

$$A_2 = \frac{4}{EI} \times \frac{4}{2} = \frac{8}{EI}$$

$$x_1 = \frac{3}{2}, \quad x_2 = \frac{2}{3} \times 4$$

Rotation

$$\begin{aligned} \text{At B: } \theta_B &= \theta_A + \theta_{BA} \quad (1^{\text{st}} \text{ MAT}) \\ &= 0 + A_1 \\ &= \frac{12}{EI} \quad (\curvearrowright) \end{aligned}$$

$$\begin{aligned}
 \text{Pt C: } Q_C &= Q_B + Q_{CB} \quad (1^{\text{st}} \text{ MAT}) \\
 &= \frac{12}{EI} + A_2 \\
 &\quad \quad \quad " 8/EI \\
 &= \frac{20}{EI} \quad (\leftarrow)
 \end{aligned}$$

Deflection (u, v)

$$v_B = 0 \quad (\text{column is axially rigid})$$

$$\begin{aligned}
 u_B &= u_A + S_{AB} (Q_{ALAB} + \bar{Q}_{BA}) \\
 " &= -1 \quad (2^{\text{nd}} \text{ MAT}) \\
 0 &
 \end{aligned}$$

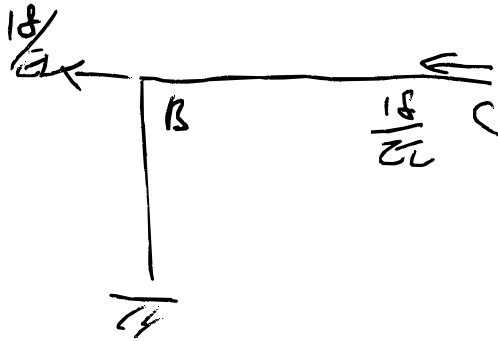
$$= -\delta_{BA}$$

$$= -A_1 \times \frac{3}{2}$$

$$= -\frac{12}{EI} \times \frac{3}{2}$$

$$= -\frac{18}{EI} \quad (\leftarrow)$$

At Pt. C



$$u_C = u_B \quad (BC \text{ is axially rigid})$$

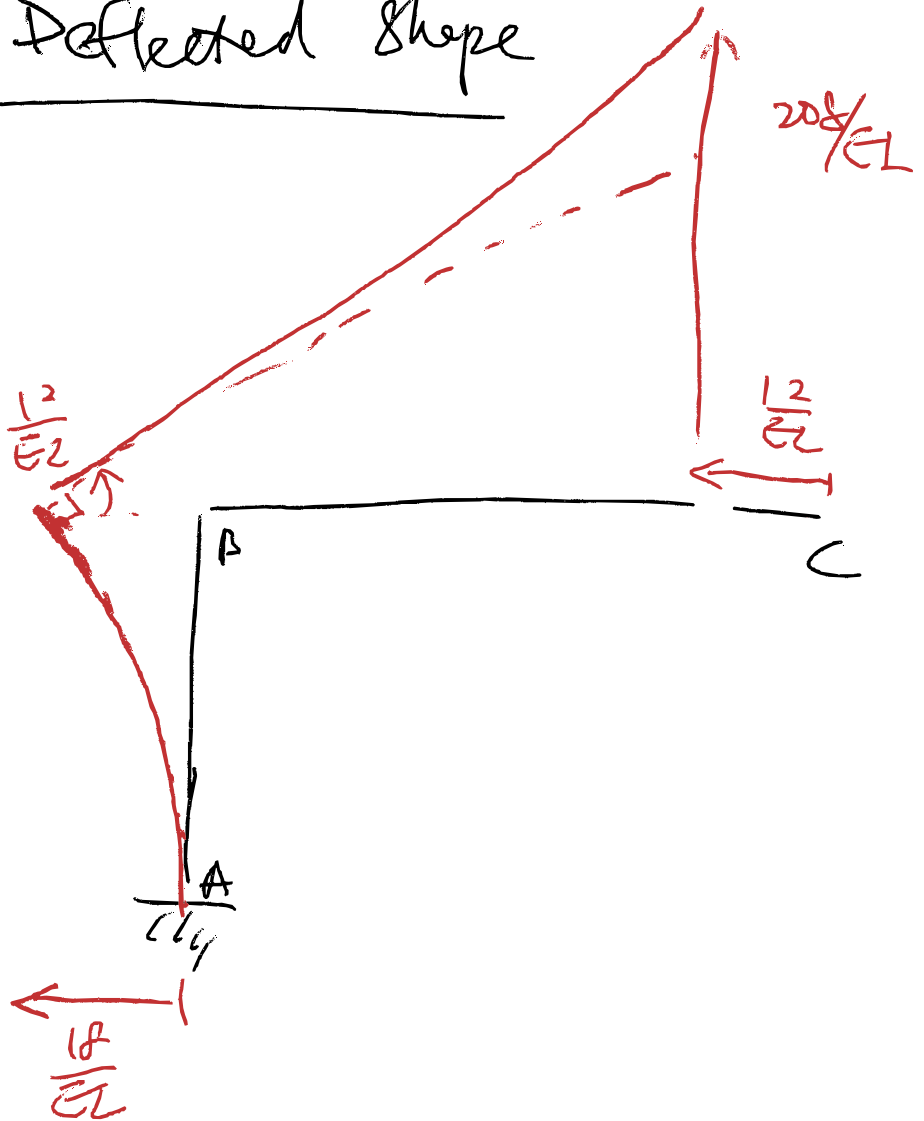
$$= -\frac{18}{EI} \quad (\leftarrow)$$

$$v_C = v_B + \theta_B L_{BC} + \delta_{CB}$$

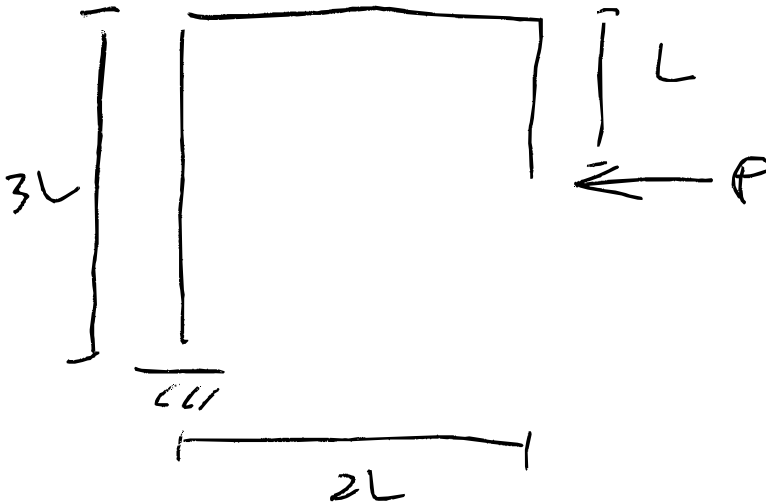
$$\begin{array}{ccccccc} \text{"} & \text{"} & \text{"} & \text{"} & & & \\ 0 & \frac{(2)}{EI} & 4 & A_2 \times \lambda_2 & & & \\ & & & \text{"} & \text{"} & & \\ & & & \frac{8}{EI} & \frac{8}{3} & & \end{array}$$

$$= \frac{208}{3EI} \quad (\uparrow)$$

Deflected shape



Example 2



load : in terms of P

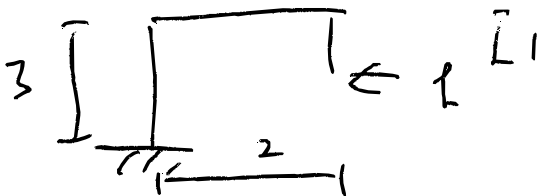
length : " " " L

M : " " " PL

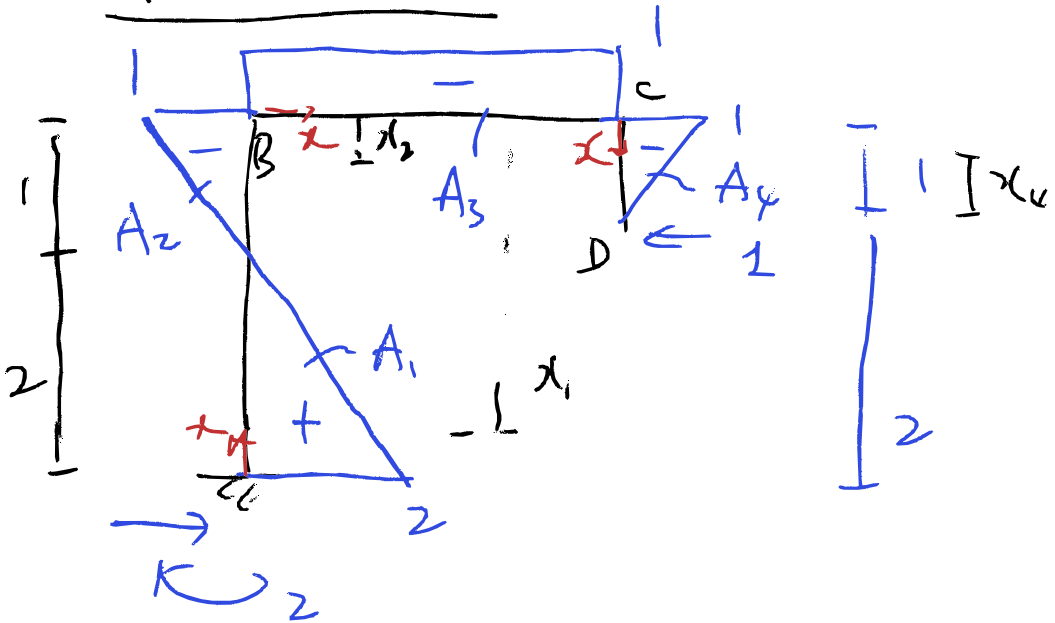
k : " " " $\frac{PL}{EL}$

Δ : " " " $\frac{PL^2}{EL}$

u, v : " " " $\frac{PL^3}{EL}$



BMP or k



$$A_1 = 2$$

$$x_1 = 1 + 2 \times \frac{2}{3} = \frac{7}{3}$$

$$A_2 = -\frac{1}{2}$$

$$x_2 = \frac{1}{3}$$

$$A_3 = -2$$

$$x_3 = 1$$

$$A_4 = -\frac{1}{2}$$

$$x_4 = \frac{2}{3}$$

PE. B

$$V_B = 0$$

$$Q_B = Q_A + Q_{BA} = \frac{3}{2} \frac{PL^2}{EI} \quad (4)$$

$\begin{matrix} \text{"} & \text{"} \\ 0 & A_1 + A_2 \end{matrix}$

$$u_B = u_A + s_{AB} (\theta_A L_{AB} + \delta_{BA})$$

0	-1	0	$A_1 x_1 + A_2 x_2$

$$= -\frac{q}{2} \frac{PL^3}{EI} (\leftarrow)$$

Pt. C

$$u_C = u_D = -\frac{q}{2} \frac{PL^3}{EI} (\leftarrow)$$

$$\theta_C = \theta_B + \theta_{CB} = -\frac{1}{2} \frac{PL^2}{EI} (\curvearrowright)$$

$\frac{3}{2}$	$A_3 = -2$	

$$V_C = V_B + \theta_B L_{BC} + \delta_{CB}$$

0	$\frac{3}{2}$	2	$A_3 x_3$
			-2
			1

$$= 1 \frac{PL^3}{EI} (\uparrow)$$

$$\frac{P_L \cdot D}{V_D = V_C = \frac{PL^3}{\epsilon L} \quad (1)}$$

$$Q_D = Q_C + Q_{DC} = -1 \frac{PL^2}{\epsilon L} \quad (2)$$

$\parallel \qquad \parallel$
 $-\frac{1}{2} \qquad A_4 = -\frac{1}{2}$

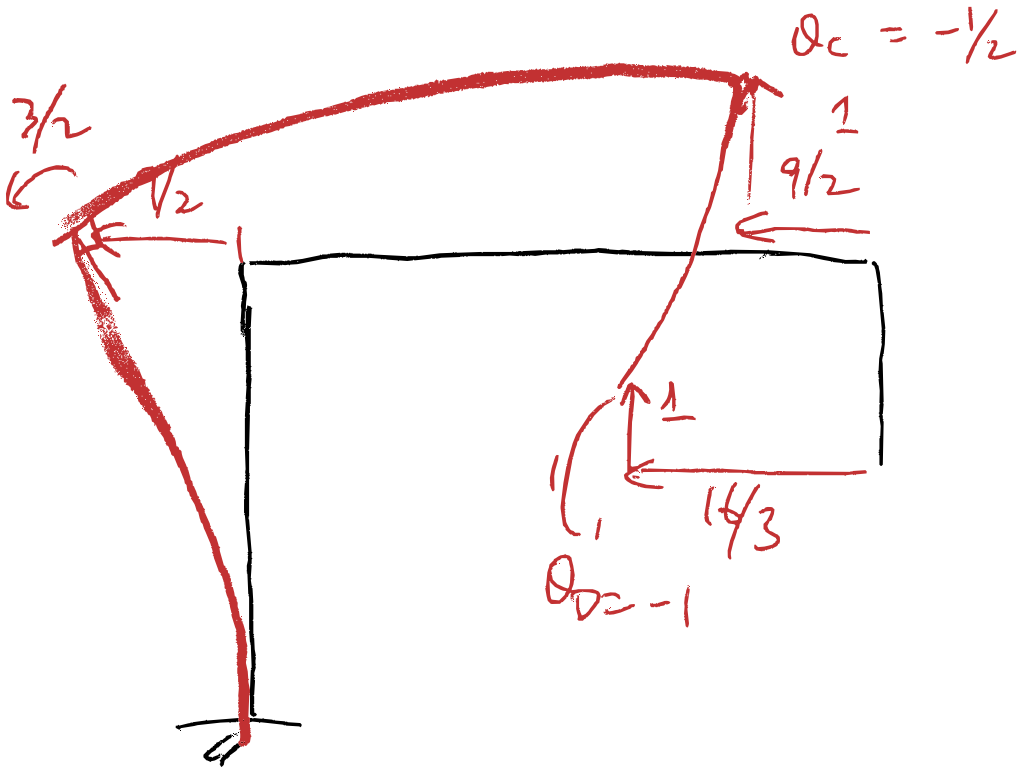
$$U_D = U_C + S_{CD}(Q_C L_{CD} + \phi_{DC})$$

$\parallel \qquad \parallel \qquad \parallel \qquad \parallel$
 $-\frac{9}{2} \qquad 1 \qquad -\frac{1}{2} \qquad 1 \qquad A_4 \lambda_4$
 $\qquad \qquad \qquad \qquad \parallel \qquad \parallel$
 $\qquad \qquad \qquad \qquad -\frac{1}{2} \qquad \frac{2}{3}$

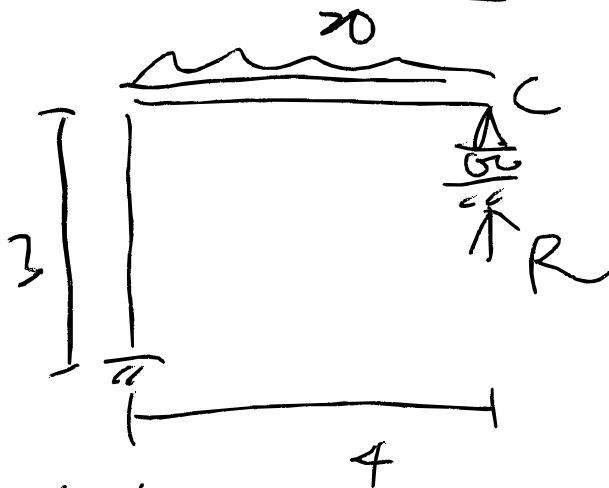
$$= -\frac{16}{3} \frac{PL^3}{\epsilon L} \quad (\leftarrow)$$

$$u:v = \frac{PL^3}{EI}$$

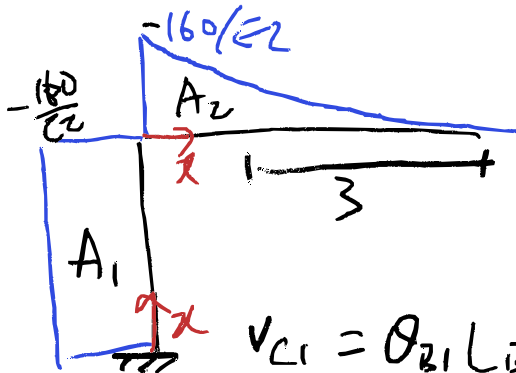
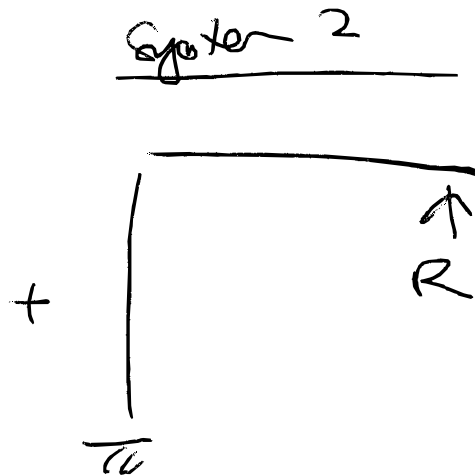
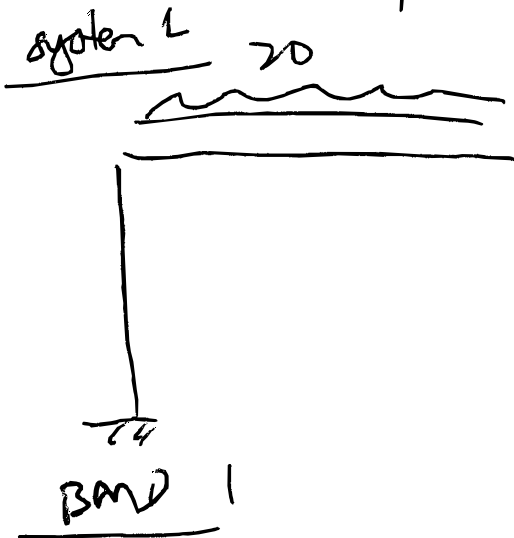
$$Q = \frac{PL^2}{EI}$$



Example 3



$$SI = 1$$



$$A_1 = -480$$

$$A_2 = -\frac{640}{3}$$

$$Q_{B1} = A_1 = -480$$

$$V_{C1} = Q_{B1} L_{BC} + \int_{CB} q dx = -2560 (\downarrow)$$

Due to R alone

$$V_{C2} = \frac{208}{3} R (\uparrow)$$

with the compatibility condition
that $V_C = 0$

$$V_{C1} + V_{C2} = 0$$

$$-2560 + \frac{208}{3} R = 0$$

$$\Rightarrow R = 36.9 (\uparrow)$$

Deflection of system L

Pt. B

$$Q_{B1} = -\frac{480}{EI}$$

$$U_{B1} = \frac{720}{EI}$$

$$V_{B1} = 0$$

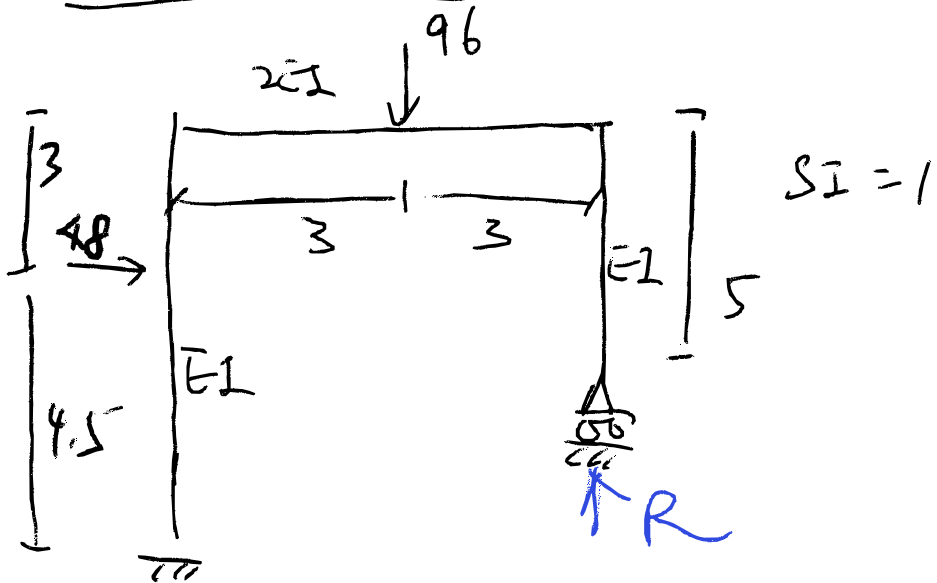
Pt. C

$$Q_{C1} = -\frac{480 - \frac{640}{3}}{EI}$$

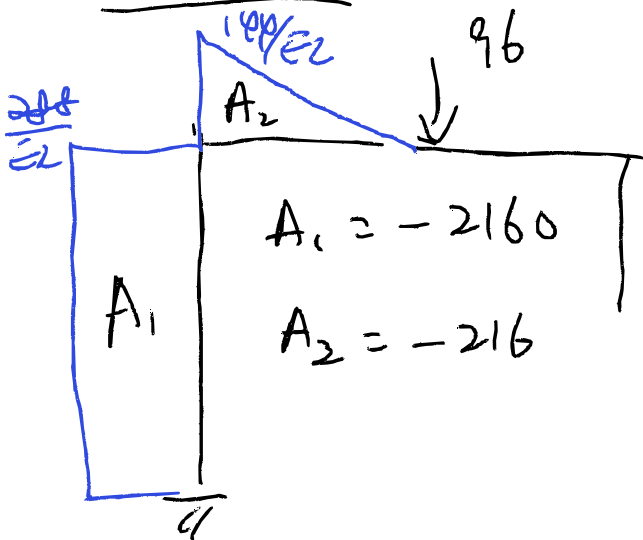
$$U_{C1} = U_{B1}$$

$$V_{C1} = -\frac{2560}{EI}$$

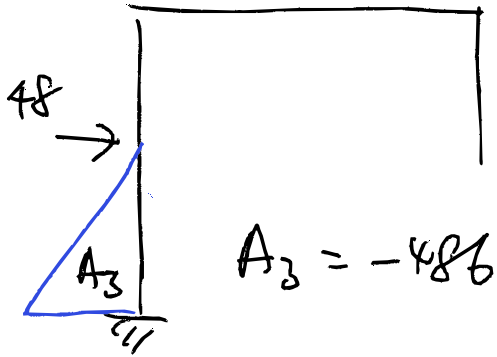
Example 4



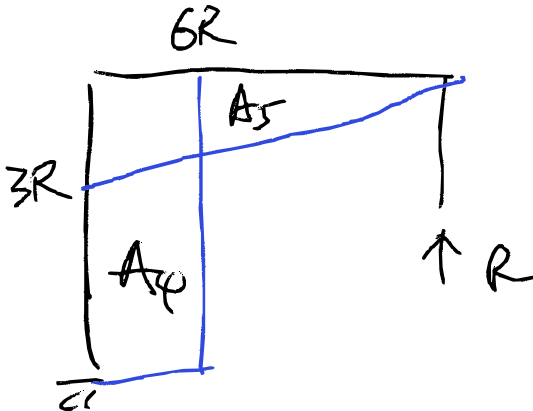
Case 1



Case 2



Case 3



$$A_4 = 45R$$

$$A_5 = 9R$$

Compatibility Condition

$$V_D = 0$$

$$\Rightarrow V_C = 0 \quad (\because V_C = V_D)$$

Case 1

$$\begin{aligned} V_{C1} &= V_{B1} + Q_{B1} L_{BC} + \delta_{CB1} \\ &\quad \begin{array}{ccc} \parallel & \parallel & \parallel \\ 0 & Q_{BA1} & A_2 \times 5 \\ & \parallel & \\ & A_1 & \end{array} \\ &= -14040 / \text{EZ} \end{aligned}$$

Case 2

$$\begin{aligned} V_{C2} &= V_{B2} + Q_{B2} L_{BC} + \delta_{CB2} \\ &\quad \begin{array}{ccc} & \parallel & \parallel \\ & Q_{BA2} & 0 \\ & \parallel & \\ & A_3 & \end{array} \\ &= -2916 \end{aligned}$$

Case 3

$$\begin{aligned} V_{C3} &= V_{B3} + Q_{B3} L_{BC} + d_{C23} \\ &\quad \quad \quad \parallel \quad \quad \parallel \quad \quad \parallel \\ &\quad \quad \quad 0 \quad \quad A_4 \quad \quad A_5 \times 4 \\ &= \frac{306R}{EI} \end{aligned}$$

$$\therefore V_{C1} + V_{C2} + V_{C3} = 0$$

$$\Rightarrow R = 55.41 (\uparrow)$$